

TADOULE LAKE, MB, Canada

WMO: 713340

Lat: 58.709N

Lon: 98.501W

Elev: 262

StdP: 98.22

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.7	(c) -36.8	(d) -43.0	(e) 0.1	(f) -38.6	(g) -40.9	(h) 0.1	(i) -36.7	(j) 8.2	(k) -21.3	(l) 7.5	(m) -23.0	(n) 2.7	(o) 300	(p) 0.780

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	26.2	15.8	24.2	15.1	22.2	14.4	17.6	23.3	16.4	21.6	15.3	20.4	3.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	11.2	19.6	14.2	10.4	18.4	13.1	9.7	17.3	50.6	23.1	47.0	21.5	43.8	20.5	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.0	(b) 7.1	(c) 6.3	(e) -39.5	(f) 29.9	(g) 0.3	(h) 1.1	(i) -39.7	(j) 30.7	(k) -39.9	(l) 31.4	(m) -40.1	(n) 32.0	(o) -40.3	(p) 32.8		
(a) 8.0	(b) 7.1	(c) 6.3	(e) -39.6	(f) 20.3	(g) 0.3	(h) 0.9	(i) -39.8	(j) 20.9	(k) -40.0	(l) 21.4	(m) -40.2	(n) 21.9	(o) -40.4	(p) 22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.1	-25.1	-24.2	-17.2	-7.1	2.1	11.4	15.5	13.4	7.3	-1.5	-14.2	-22.7
	DBStd	15.85	7.26	6.52	7.73	7.77	5.80	4.81	3.49	3.43	4.27	4.78	7.67	7.52
	HDD10.0	5904	1088	958	843	513	252	41	3	8	103	358	724	1014
	HDD18.3	8566	1346	1191	1101	763	502	212	99	158	330	616	974	1272
	CDD10.0	402	0	0	0	0	8	83	175	113	23	1	0	0
	CDD18.3	22	0	0	0	0	0	4	13	5	0	0	0	0
	CDH23.3	257	0	0	0	0	6	67	124	57	3	0	0	0
	CDH26.7	34	0	0	0	0	1	8	18	7	0	0	0	0
Wind	WSAvg	3.2	3.1	3.2	3.2	3.3	3.3	3.3	3.2	3.2	3.5	3.5	3.1	3.0
Precipitation	PrecAvg	428	15	13	20	18	30	51	74	67	61	43	21	14
	PrecMax	657	35	34	55	39	63	85	141	132	112	81	48	32
	PrecMin	326	5	3	5	5	6	18	37	26	21	16	6	4
	PrecStd	68	7	7	12	9	15	17	25	26	21	17	10	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.9	-3.9	6.2	13.2	23.6	27.7	29.1	27.7	22.4	14.2	4.4	-3.8
		MCWB	-3.3	-5.3	2.0	6.9	12.6	15.0	18.4	18.1	14.5	8.6	2.0	-4.7
	2%	DB	-6.6	-8.2	2.5	9.0	19.1	25.4	26.3	24.9	18.5	9.7	1.0	-5.6
		MCWB	-7.0	-8.9	-0.4	4.0	10.2	14.3	16.1	16.5	12.6	6.5	0.1	-5.9
	5%	DB	-10.1	-10.6	-0.7	6.2	15.5	23.0	24.5	22.1	16.0	6.8	-1.3	-8.2
		MCWB	-10.5	-11.3	-2.4	2.3	8.8	13.3	15.6	15.4	11.4	5.0	-1.9	-8.5
	10%	DB	-13.9	-13.3	-4.5	3.9	12.0	20.7	22.6	19.9	13.8	4.6	-3.1	-11.3
		MCWB	-14.2	-13.8	-5.7	1.0	6.8	12.7	15.0	14.3	10.2	3.3	-3.6	-11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.2	-5.1	2.1	7.5	13.7	16.9	19.5	19.0	15.2	8.9	2.3	-4.4
		MCDB	-3.0	-3.7	5.8	12.8	21.5	23.7	25.7	26.1	21.2	12.0	3.8	-4.0
	2%	WB	-7.0	-9.0	-0.2	4.4	11.0	15.5	17.9	17.3	13.5	6.9	0.1	-6.0
		MCDB	-6.7	-8.3	2.1	8.4	18.8	22.4	23.5	23.2	17.3	9.5	0.9	-5.5
	5%	WB	-10.6	-11.3	-2.5	2.5	9.0	14.2	16.8	16.1	12.0	4.9	-1.8	-8.6
		MCDB	-10.2	-10.6	-0.6	5.7	15.5	21.1	22.0	20.9	15.3	6.6	-1.3	-8.3
	10%	WB	-14.3	-13.8	-5.6	1.1	7.1	13.2	15.8	14.8	10.6	3.4	-3.6	-11.7
		MCDB	-13.9	-13.2	-4.4	3.7	11.6	20.0	20.6	18.7	13.5	4.6	-3.1	-11.4
Mean Daily Temperature Range	5% DB	MDBR	8.6	10.3	12.6	12.1	11.0	11.7	10.2	9.1	7.4	5.4	7.6	7.8
		MCDBR	11.2	12.5	15.3	12.9	17.3	15.8	13.5	13.3	11.0	8.8	7.5	9.3
		MCWBR	11.1	11.8	12.8	8.8	10.0	7.1	5.8	6.6	6.2	6.1	7.0	9.1
	5% WB	MCDBR	11.2	12.4	14.5	12.2	16.4	13.6	11.2	12.0	10.1	7.7	7.2	9.3
		MCWBR	11.0	11.7	12.2	8.5	9.7	6.9	5.7	6.4	6.3	5.6	6.7	9.2
Clear-Sky Solar Irradiance	taub	0.213	0.234	0.257	0.343	0.333	0.342	0.386	0.359	0.311	0.273	0.236	0.185	
	taud	2.145	2.269	2.265	2.058	2.327	2.433	2.354	2.436	2.548	2.471	2.119	2.065	
	Ebn at Noon	647	826	902	853	898	893	839	840	839	771	580	553	
	Edh at Noon	45	67	92	137	113	104	110	95	72	56	46	33	
All-Sky Solar Radiation	RadAvg	0.60	1.61	3.35	5.18	5.83	5.76	5.23	4.01	2.55	1.21	0.57	0.35	
	RadStd	0.04	0.11	0.22	0.42	0.47	0.57	0.36	0.32	0.24	0.13	0.07	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page